

MMMC: A Massively Multi-way-aligned Multilingual Corpus¹

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Abstract

Much of the bilingual data used to train large Neural Machine Translation (NMT) models comes from noisy sources scraped from the internet and is vastly dominated by high-resource languages such as English, Chinese, or Spanish. We release a parallel text corpus, called the Massively Multi-way-aligned Multilingual Corpus (MMMC),¹ consisting of pairs of English source segments and target translations in 98 languages, with the unique characteristic that a large majority of the translations are from the same source, providing the possibility to pivot through the English to create data for 9,702 translation directions. Many of the target languages are low-resource, including Austronesian languages such as Chuukese, Kosraean, and Pohnpeian, which currently have no data available on OPUS.² This corpus contains data sourced from the translation memories created by translators working for The Church of Jesus Christ of Latter-day Saints. We showcase the quality of this corpus through experiments finetuning the No Language Left Behind (NLLB) model, achieving up to a 10.2 chrF++ score improvement for certain language directions even in an out-of-domain scenario. We plan to release this corpus under an ODC license or similar license.^{3,4}

1 Introduction

Machine Translation (MT) is an important tool for all people in the world. The vast majority of languages spoken around the world are underrepresented in Natural Language Processing (NLP) and MT research. The current state-of-the-art Neural Machine Translation (NMT) models require vast amounts of source and target aligned translation

data to produce high quality translations, despite efforts to mitigate this requirement using back translation of monolingual data or a variety of unsupervised learning strategies.

For high-resource languages, there is much support because of the amount of parallel data for those languages, but high-resource languages constitute only a small fraction of the world’s languages. In contrast, the vast majority are low-resource, lacking adequate support, and making progress in NMT and NLP technologies for these languages difficult and sometimes almost impossible.

We introduce the Massively Multi-way-aligned Multilingual Corpus (MMMC),¹ which includes data in 98 languages (not including English). We source this high-quality parallel text data directly from the translation memories created by translators working for The Church of Jesus Christ of Latter-day Saints.⁵ We perform extensive cleaning to ensure the quality of each language pair within the dataset. We acknowledge that these bilingual pairs are *translationese* (Graham et al., 2019), but they can still be tremendously valuable in training multilingual models, as demonstrated by the experiments in this work. Future updates to the corpus are anticipated as more content is translated in these and additional languages. In total, we contribute:

- A parallel text corpus of approximately 46 million parallel segments, supporting 99 languages, which will receive future updates to include additional high-quality parallel translation segment pairs.
- Parallel translation data for Pacific Island, Creole, indigenous American, and African languages for which parallel data does not exist elsewhere to our knowledge.
- Empirical evidence demonstrating that utilizing the data in this corpus enhances MT system performance, as shown by finetuning

¹The name has been anonymized and may change in the final version of the paper.

²<https://opus.nlpl.eu/>

³This corpus will be released at the time of publication.

⁴<https://opendatacommons.org/licenses/>

⁵<https://www.churchofjesuschrist.org>

NLLB and achieving 0.5 to 10.2 chrF++ gains in 57.7% of 6,320 language directions existing in NLLB.

- To our knowledge, the first open-weight MT system supporting Chuukese, Kosraean, Pohnpeian, and Tahitian trained on human-quality parallel data, released with this work.⁶
- Empirical evidence demonstrating the utility of this corpus even in bilingual, single language pair finetuning scenarios, when compared to using data from Paracrawl (Bañón et al., 2020).

2 Related Works

There are many parallel text corpora available on the internet for the purpose of training MT and other large language models. Utilization of religious material for creating parallel text corpora has been fruitful in creating basic MT systems for low-resource languages (Agić and Vulić, 2019). Most notably, McCarthy et al. (2020) align parallel bible translations in over 1600 languages. These corpora are extensive in language coverage, although they do not include diverse genres that are more valuable for the usability of general MT systems.

Much larger parallel text corpora also exist, but are typically created using automatic alignment methods. Creators of these corpora utilize internet crawlers, which identify websites containing translations of content and then use various methods for document and sentence alignment to obtain parallel translation pairs (Resnik and Smith, 2003; Munteanu and Marcu, 2005; Koehn, 2005; Bañón et al., 2020). Some of these corpora are high quality and contain many multi-way aligned segments, such as Europarl⁷ which currently includes 21 languages in 420 possible translation directions.

Techniques for aligning non-parallel monolingual data, such as the use of language identification systems and sentence embeddings, have also been employed to create extremely large, though somewhat noisy, bitext corpora to train large multilingual NMT models like NLLB (Team et al., 2022).

Much effort has been made to curate datasets utilizing the vast amount of parallel data on the internet. Most notable is OPUS (Tiedemann, 2012), an internet repository containing over 1000 parallel text corpora that can be sorted and accessed by

⁶We will include links to this in the final version of the paper.

⁷<https://opus.nlpl.eu/Europarl/corpus/version/Europarl>

English → X size	Languages
1,000,000+	15
500,000-1,000,000	22
100,000-500,000	23
50,000-100,000	25
10,000-50,000	10
1-10,000	3

Table 1: Distribution of the number of segments in the dataset per language.

language pair. Similarly, (Gowda et al., 2021) have released tools for processing parallel text corpora.

Other researchers have focused on curating parallel corpora for specific language pairs, usually of low-resource languages. (Robinson et al., 2024) create a parallel text corpora for 41 Creole languages in 172 translation directions by searching for books with translations and contacting researchers within Creole language communities for potential data sources. In addition, they utilize data repositories such as MTData (Gowda et al., 2021), OPUS (Tiedemann, 2012), and LDC.⁸ The authors found that this much more deliberate searching of data proved very effective in improving the performance of MT systems for those Creole languages.

Higher quality multilingual parallel corpora have also been created for the purpose of training MT models by involving human reviewers. Team et al. (2022) have created both the FLORES benchmark dataset and the NLLB Seed dataset. The FLORES translation workflow included a multi-step process that involved translators, automatic checkers, human reviewers, and post-editors to create a high-quality translation dataset free of noise from unaligned segments. The NLLB Seed dataset was professionally translated by humans so that the automatic methods could be trained using this high quality "seed" data.

In NMT models, low-resource language pairs are more sensitive to noisy data than statistical MT models (Belinkov and Bisk, 2017; Koehn et al., 2018). Because of this, parallel corpora need to be cleaned thoroughly to improve the data quality, which then results in improved NMT model output (Team et al., 2022; Robinson et al., 2024; Minh-Cong et al., 2023).

	fra	kek	khm	kor	lin	mah	pes	tel	tur	ukr
fra		155.5K	601.5K	1.16M	92.2K	283.1K	77.7K	64.8K	100.6K	779.3K
kek	155.5K		125.7K	135.2K	71.8K	147.3K	63.9K	58.3K	57.5K	149.1K
khm	601.5K	125.7K		541.8K	73.2K	250.6K	64.0K	62.9K	74.5K	495.6K
kor	1.16M	135.2K	541.8K		78.6K	250.6K	73.0K	59.5K	86.8K	662.6K
lin	92.2K	71.8K	73.2K	78.6K		89.8K	42.7K	45.0K	64.1K	89.4K
mah	283.1K	147.3K	250.6K	250.6K	89.8K		64.0K	62.0K	76.2K	257.4K
pes	77.7K	63.9K	64.0K	73.0K	42.7K	64.0K		49.4K	53.2K	72.8K
tel	64.8K	58.3K	62.9K	59.5K	45.0K	62.0K	49.4K		38.3K	60.5K
tur	100.6K	57.5K	74.5K	86.8K	64.1K	76.2K	53.2K	38.3K		92.2K
ukr	779.3K	149.1K	495.6K	662.6K	89.4K	257.4K	72.8K	60.5K	92.2K	

Figure 1: The number of parallel segments for 10 multi-way aligned languages when pivoted through English.

3 Dataset

MMMC contains approximately 46 million total parallel translation segments across the 98 English $\rightarrow$ X language directions. The language pair with the most data, English $\rightarrow$ Spanish, includes about 2.3M parallel segments while the language with the least data is English $\rightarrow$ Kazakh which only includes 5K parallel segments. Table 1 shows the distribution of the number of English $\rightarrow$ X segments across all the included languages in the dataset.

Refer to Table 7 in Appendix B for details on each language in the corpus and the number of segments aligned with English.

A significant contribution of this corpus is the inclusion of languages within the Austronesian language family, typically spoken in the island countries of the Pacific Ocean. 18 Austronesian languages are included: Cebuano, Chuukese, Fijian, Kiribati, Hiligaynon, Ilokano, Indonesian, Kosraean, Marshallese, Malagasy, Malay, Palauan, Pohnpeian, Samoan, Tahitian, Tagalog, Tongan, and Yapese.

Uniquely, parallel text data for three of the Austronesian languages (Chuukese, Kosraean, and Pohnpeian) is not found on OPUS. The corpus also contains over 400K parallel segments for Tahitian and over 100K parallel segment pairs for Hmong, while OPUS contains fewer than 100 parallel segment pairs for both those languages.

3.1 Multi-way Alignment

Freitag and Firat (2020) have shown that training complete Multilingual Neural Machine Translation (cMNMT) systems, which include parallel data to and from every language represented in the data, yields impressive translation results. The

unique overlap within the English source segments of MMC has the potential to significantly enhance massively multilingual MT systems.

We utilize the overlap across the 98 target languages to create the multi-way aligned corpus. Using English as a pivot, we can create an additional 9,506 unique $X \rightarrow Y$ language pairs. Including English $\rightarrow$ X and $X \rightarrow$ English, the corpus supports 9,702 language directions, making it one of the largest multi-way aligned corpora by number of supported language directions. On average, there are 156 thousand parallel segment pairs for each language direction, totaling 1.5 billion overall. We reference a heat map visualizing the number of parallel segments per language pair in Appendix B.

To show an example of the alignment between 10 diverse non-English Languages, we provide Figure 1, which includes the number of aligned segments between the following languages: French, Kekchi, Khmer, Korean, Lingala, Marshallese, Persian, Telugu, Turkish, and Ukrainian. Even within this diverse set of languages, the average number of parallel segments is 179K for all 90 possible language directions.

Table 2 depicts two segments from the dataset with their translations across eight different languages. We point out the high-quality alignments between higher-resource Hindi, Japanese, Spanish, Urdu, and Bulgarian sentences to the lower-resource Lao, Navajo, and Twi segments.

3.2 Dataset Sources

All English source segments and corresponding translations were extracted from translation memories created by translators working for the Church of Jesus Christ of Latter-day Saints. Despite the religious nature of these sentences, there is still a variety of genres in the dataset, including, but not

⁸<https://www ldc.upenn.edu/>

Language	Translation
English	As members of God's family, we are truly brothers and sisters.
Bulgarian	Като членове на Божието семейство, ние действително сме братя и сестри.
Hindi	परमेश्वर के परिवार के सदस्य होने के नाते, हम वास्तव में भाई-बहन हैं।
Japanese	神の家族として、わたしたちは真の兄弟姉妹です。
Lao	ໃນ ຖ້າ ນະ ສະ ມາ ຊິກ ໃນ ຄອບ ຄົວ ຂອງ ພຣະ ເຈົ້າ, ເຮົາ ເປັນ ອ້າຍ ເຮືອນ ນ້ອງ ກັນ ແທ້ໆ.
Navajo	Di Yin God bighan haz'áá'dóó da'niitahgo, t'áá ániiltso ahidadiilchíín.
Spanish	Como miembros de la familia de Dios, verdaderamente somos hermanos y hermanas.
Twi	Se Onyankopon abusua mu mma no, nokore mu yeye anuanom mmарima ne mmaa.
Urdu	خدا کے خاندان کے ارکان کی حیثیت سے، ہم سب واقعی بھائی اور بہنیں ہیں۔
English	These beautiful animals can accelerate from a standstill position to running as fast as 68 mph (109 km/h) in less than three seconds.
Bulgarian	Тези красиви животни могат да ускоряват от неподвижно положение до скорост от 109 км/ч за по-малко от три секунди.
Hindi	इन खूबसूरत जानवरों की गति तीन सेकंड से भी कम समय में 109 कि.मी. तेज गति तक पहुँच जाते हैं।
Japanese	この美しい動物は、静止状態から3秒以内に、時速109キロの速さまで加速できるのです。
Lao	ສັດທີ່ສວຍງາມເຫລົ່ານີ້ສາມາດເລ່ນໄວຈາກທຳທີ່ຢືນເຖິງ 68 ໄມ (109 ກິໂລແມັດຕໍ່ຊົ່ວໂມງ) ພາຍ ໃນສາມວິນາທີ.
Navajo	Diidi naaldloosh danizhóní t'óó siziídoó 68 tsin sitá biighahgo dah diilwo' mph (109 km/h) táá' dah álzhin t'áá bich'j'óo.
Spanish	Esos hermosos animales pueden acelerar desde el punto de partida hasta una velocidad de 109 km/h en menos de tres segundos.
Twi	Saa mmoa ahoofefoo yi gyina faako a, otumi hyeasee tu mmirika twa akwansini 68 wo donhwere baako wo mmere kekeke mmiensa ntam.
Urdu	یہ خوب صورت جانور ساکت حالت سے ۶۸ میل فی گھنٹہ (۱۰۹ کلو میٹر فی گھنٹہ) کی رفتار صرف تین سیکنڈ سے کم وقت میں پکڑ سکتا ہے۔

Table 2: Two example segments from the dataset with their translations in a variety of languages.

limited to:

- Religious instruction and discourses, which include a variety of stories, analogies, and modern speech
- Scripture
- Humanitarian, emergency response, welfare, and community service materials
- Family History resources
- Administrative documents and instruction

3.3 Cleaning

As the data in the corpus was taken from translation memories, it originally included items commonly found in such data and has required additional cleaning that would not typically be performed on data from other sources. We created an extensive cleaning pipeline that follows several guidelines found in the GILT Leaders Forum’s Best Practices in Translation Memory Management.⁹ Due to the English-centric nature of the dataset, this pipeline was designed with the assumption that English is the source language. We optimize our cleaning steps to remove noise over preserving potentially good data. Appendix A includes the cleaning steps applied to all languages.

⁹<https://github.com/GILT-Forum/TM-Mgmt-Best-Practices/blob/master/best-practices.md>

In addition, many automatic and manual passes through the data were performed to identify any confidential or potentially sensitive content and to remove it from the released version of the corpus.

4 Experiments

Though the number of parallel segments between many languages is small compared to other corpora, we showcase their quality and impact through three main experiments. We first finetune the NLLB model (Team et al., 2022) using the 80 languages which overlap with the 200 previously supported languages and evaluate on the standard FLORES dev and devtest sets.¹⁰ Second, we take the remaining 18 languages which are not supported by FLORES or NLLB and we finetune the NLLB model while evaluating on our own held out evaluation sets, creating the first MT system to our knowledge that contains human-quality parallel data for 5 languages: Chuukese, Kosraean, Pohnpeian, Tahitian, and Navajo. Finally, we finetune NLLB in a bilingual setting, comparing to using bilingual data from Paracrawl (Bañón et al., 2020), showcasing the value of MMC in a non-multilingual scenario.

For all experiments, we finetune nllb-200-

¹⁰https://huggingface.co/datasets/openlanguage/flores_plus

distilled-1.3B^{11,12} on all language directions, utilizing the complete alignment of our corpus. All models are trained to convergence with a learning rate of 1e-4 using 8 A100 GPUs with a batch size of 4096 translation pairs. Inference is performed on all possible directions for each model on held-out test sets.

4.1 Experiment 1: 80 FLORES Supported Languages

To show the benefit of the high-quality data included in MMMC, we evaluate the effect of finetuning NLLB on 80 of the languages of this corpus that are included in the FLORES evaluation data. We choose this subset of the corpus to avoid the need to modify the NLLB tokenizer and to be able to finetune on and evaluate all 6320 language directions using FLORES. It is important to acknowledge that this task is particularly challenging due to the domain mismatch between the religious training data and the FLORES evaluation data.

We compare our finetuned model to the base 1.3B parameter NLLB model keeping the same tokenizer. The model is finetuned for the 80 different languages using a temperature of 5.0 to sample the different language directions. We validate every 256 steps and use the FLORES dev set to determine convergence. For inference, the model generates hypotheses using beam search with a beam width of 5.

See Table 7 in Appendix B for the list of the languages with FLORES support, all of which are included in this model. We evaluate the models using spBLEU (Goyal et al., 2022) with the FLORES-200 tokenizer and chrF++ (Popović, 2017) using SacreBLEU (Post, 2018).

4.2 Experiment 2: 18 Previously Unsupported Languages

For the remaining 18 languages unsupported by NLLB and FLORES, we create a separate model that can translate between all 18 of these languages in addition to English. This model supports the following languages: Bislama, Chuukese, Efik, Fante, Kiribati, Hindi (Fiji), Hiligaynon, Hmong, Kekchi, Kosraean, Marshallese, Navajo, Palauan, Pohnpeian, Quichua (Ecuador), Tahitian, Tongan, and Yapese. This is a diverse assortment of languages,

¹¹<https://huggingface.co/facebook/nllb-200-distilled-1.3B>

¹²We choose this model over the larger counterparts due to limited training resources.

Language Pair	Paracrawl	MMMC
eng_Latn → kor_Hang	100K	100K
eng_Latn → lvs_Latn	100K	100K
eng_Latn → mya_Mymr	31.4K	19.6K
eng_Latn → swh_Latn	132K	132K

Table 3: Number of sentence pairs sampled from the Paracrawl and MMMC corpora for Experiment 3 (Section 4.3). All sentence pairs are used from both corpora for Burmese (mya) and all sentence pairs are used from Paracrawl for Swahili (swh).

and thus further research could be conducted to analyze the positive and negative transfer between the related and unrelated languages in this dataset. We simply train on the data for this group of languages and report our results.

We created validation and test datasets using the common sentences across all 18 languages except for Quichua. As can be seen in Figure 3 in Appendix B, Quichua (que) has little overlap with other lower-resource languages, which comprise the majority of these 18 languages, and so we elected to only evaluate Quichua to and from English, while evaluating all other directions for the other languages. For the 17 other languages, we sampled 2000 of the completely aligned segments using English as the pivot language to create the 1000 segment validation and 1000 segment test sets for all 306 language directions. This validation set is used to determine convergence, validating every 256 steps.

Like the previous experiment, generation is done using beam search with a beam width of 5. When batching, we sample from the possible language directions using a higher temperature of 50.0 so that the lower resource languages can be seen more often by the model. We do not alter the NLLB tokenizer in order to encourage more knowledge transfer.

Because none of these languages are supported by NLLB, we introduce new language indicator tokens, expanding the vocabulary of the model by a small amount. We evaluate the model using BLEU (Papineni et al., 2002) and chrF++ (Popović, 2017) using SacreBLEU (Post, 2018).

4.3 Experiment 3: Paracrawl Corpus Comparisons

To further showcase the quality of the MMMC corpus outside of a multilingual scenario, we conduct

small-scale experiments comparing finetuning of single language pairs in MMC to those included in the Paracrawl (Bañón et al., 2020) corpus.

We select four language pairs, covering a variety of resource-levels and scripts: English → Korean (kor), English → Latvian (lvs), English → Burmese (mya), and English → Swahili (swl). For training, sentence pairs are randomly sampled from each corpus, with the number of sampled sentence pairs shown in Table 3. All available Swahili and Burmese Paracrawl data and all available Burmese MMC data are used in these experiments. To better reflect a realistic low-resource scenario, we choose to use all Burmese Paracrawl data for training despite it being larger than the MMC Burmese data. We use the FLORES dev and devtest datasets for validation and testing.

For each language pair, we finetune the 1.3B Parameter NLLB model using the sampled Paracrawl data and compare its performance on the held-out FLORES devtest set to that of NLLB finetuned with the sampled MMC data. We evaluate all eight models using BLEU (Papineni et al., 2002), chrF++ (Popović, 2017), and COMET (Rei et al., 2020). When calculating BLEU scores, we tokenize the Korean using the ko-mecab tokenizer and the Burmese using the flores-200 tokenizer. We use the default COMET model, Unbabel/wmt22-comet-da (Rei et al., 2022).

5 Results and Discussion

5.1 Experiment 1: 80 FLORES Supported Languages

After finetuning NLLB on the 80 supported languages, we evaluated all 6320 language directions using the FLORES devtest set. Aggregated scores based on resource amounts and whether or not English is included in the language pair are shown in Table 4. These results compare the baseline NLLB model to the finetuned counterpart on all 80 languages and 6320 possible language directions.

When translating from English to a low-resource language, we see that there are improvements in chrF++ scores but there is a decrease in performance according to the BLEU score (-0.24 BLEU) on average. Meanwhile, when going from a low-resource language into English, we see modest improvements according to both metrics. Going from English to a high resource language, we also see performance gains after finetuning on the MMC corpus, while going from High resource into En-

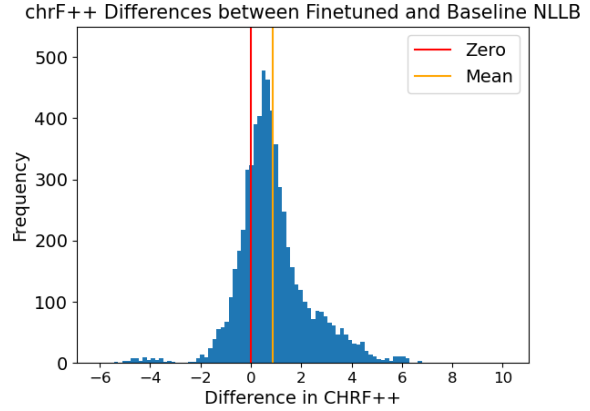


Figure 2: Histogram showing the differences between the chrF++ scores for the baseline and finetuned models. Positive values indicate a higher chrF++ score for the finetuned model. Zero difference and the mean difference are indicated by the red and orange lines respectively.

glish, the baseline NLLB model does slightly better.

Focusing on chrF++ scores, we see the greatest improvements when a LRL (Low Resource Language) is the target with a 1.29 chrF++ score increase after finetuning for Low → Low and a 0.97 score improvement for High → Low.

The high resource to high resource X → Y language pairs show improvements after finetuning on MMC, which we believe is due to NLLB’s training data being primarily English centric with X → Y translation pairs included for only a relatively small number of the 200 languages included in the model, while our dataset has high amounts of overlap for most X → Y translation pairs. These results are consistent with previous work by Freitag and Firat (2020).

The overall mean spBLEU and chrF++ increased by 0.56 and 0.87 respectively. While this improvement is modest, it is important to note that the data used to finetune NLLB differs in domain from the test set used in this evaluation. Notably, these gains persist despite the domain differences.

We show the differences between chrF++ scores for the baseline and finetuned models across all language directions in Figure 2. Approximately 77% (4868/6320) of directions either showed improvement after finetuning or did not change at all and about 57.7% (3645/6320) achieved at least a 0.5 chrF++ score improvement. In Table 5, we highlight specific language pairs with the most positive chrF++ deltas and with the 10 most negative deltas

Language Pair	Baseline		Finetuned	
	BLEU	chrF++	BLEU	chrF++
English → Low Resource	27.24	45.79	27.00	46.56
Low Resource → English	36.45	56.19	36.51	56.55
English → High Resource	36.22	53.57	36.99	54.17
High Resource → English	42.59	62.11	42.36	62.03
Low Resource → Low Resource	17.42	36.15	17.78	37.44
Low Resource → High Resource	21.96	40.61	22.78	41.52
High Resource → Low Resource	19.96	39.05	20.13	40.02
High Resource → High Resource	26.17	45.02	26.90	45.62
Overall Mean	22.26	41.11	22.82	41.98

Table 4: Aggregated BLEU and chrF++ score results comparing the base 1.3B parameter NLLB to the finetuned version using the Massively Multi-way-aligned Multilingual Corpus on the FLORES devtest set. Low resource and high resource categories are based on classifications by Team et al. (2022). This task is particularly challenging due to the domain mismatch between the training data and the FLORES evaluation benchmark, which may impact model performance.

Rank	Language Direction	Δ chrF++
1	zho_Hant -> lvs_Latn	10.2161
2	twi_Latn -> lvs_Latn	8.3779
3	tpi_Latn -> mya_Mymr	7.4756
4	kam_Latn -> plt_Latn	7.3060
5	nob_Latn -> mya_Mymr	7.0017
6	hin_Deva -> mya_Mymr	6.7763
7	ilo_Latn -> mya_Mymr	6.7463
8	ceb_Latn -> mya_Mymr	6.6832
9	afr_Latn -> mya_Mymr	6.6487
10	sot_Latn -> mya_Mymr	6.4815
	:	
6311	ron_Latn -> lao_Lao	-4.9239
6312	zsm_Latn -> lao_Lao	-4.9517
6313	ita_Latn -> lao_Lao	-5.0660
6314	deu_Latn -> lao_Lao	-5.0911
6315	ceb_Latn -> lao_Lao	-5.2598
6316	fra_Latn -> lao_Lao	-5.3727
6317	tgl_Latn -> lao_Lao	-5.4207
6318	por_Latn -> lao_Lao	-5.5063
6319	mlt_Latn -> lao_Lao	-5.6567
6320	eng_Latn -> lao_Lao	-6.0931
Mean		0.8719

Table 5: Language pairs with the highest chrF++ Δ change after finetuning NLLB using the Massively Multi-way-aligned Multilingual Corpus. We show the the 10 most positive chrF++ deltas and with the 10 most negative deltas after finetuning, including each overall ranking.

after finetuning. The full ranking will be included with the release of the corpus. A histogram of the BLEU score improvements is included in Figure 4 in Appendix C.

The 73 lowest ranked language directions include Lao (lao_Lao) as the target language. We hypothesize that this is due to the domain mismatch of the Lao data in our corpus compared to the quality Lao data of a more general domain in the NLLB training data. The Lao data in MMMC is primarily made of religious instruction manuals and scripture, whereas the NLLB training data includes far more general sentences. Meanwhile, 41 of the top 50 improvements include Burmese (mya_Mymr) as the target language. Our corpus contains only about 20K parallel English-Burmese segments despite large improvements upon the baseline NLLB model. We hypothesize this is due to poor quality training data in NLLB for Burmese. Future studies should further investigate these phenomena.

5.2 Experiment 2: 18 Previously Unsupported Languages

In our second experiment, we finetuned NLLB on 18 new languages previously unsupported by the model and also unsupported by the FLORES evaluation set. Table 8 in Appendix C shows aggregated BLEU and chrF++ scores by source and target language, not including Quichua (que). We choose not to compare to the base NLLB model due to the lack of support for any of these languages. These promising scores showcase the value of this corpus when performing in-domain translations. We

Language Pair	Paracrawl			MMMC		
	BLEU	chrF++	COMET	BLEU	chrF++	COMET
eng_Latn → kor_Hang	27.73	33.06	0.8684	27.30	32.42*	0.8880*
eng_Latn → lvs_Latn	24.35	51.69	0.8551	24.10	52.36*	0.8706*
eng_Latn → mya_Mymr	9.10	23.10	0.6597	18.93*	33.85*	0.8804*
eng_Latn → swh_Latn	28.42	54.47	0.7709	35.88*	62.16*	0.8521*

Table 6: Performance (BLEU, chrF++, COMET) of NLLB models fine-tuned on Paracrawl and MMC datasets for four language pairs: English to Korean, Latvian, Burmese, and Swahili. Statistically significant ($p < 0.05$) results are denoted with an asterisk (*).

report BLEU scores for all directions evaluated in Table 9 in Appendix C, which includes Quichua results not shown in Table 8.

We will provide these model weights and evaluation sets with the release of the corpus.

5.3 Experiment 3: Paracrawl Corpus Comparisons

After finetuning NLLB on Paracrawl and MMC for four language pairs, we evaluated the models using the FLORES devtest set. Results are shown in Table 6.

For the higher-resource Korean data, the model finetuned on Paracrawl achieved higher BLEU and chrF++ scores, while the model finetuned on the MMC corpus obtained a statistically significantly higher COMET score, though all metrics were relatively close. The results for Latvian followed a similar pattern: BLEU and chrF++ scores were comparable across both datasets, but the MMC-finetuned model produced a larger, statistically significant improvement in COMET.

For the lower-resource language pairs, there was a substantial quality gap between the Paracrawl and MMC finetuned models, with all differences reaching statistical significance. Burmese saw a 9.83 BLEU improvement and Swahili a 7.46 BLEU improvement, with chrF++ and COMET scores showing similar gains. Notably, all available Paracrawl data was used for these two languages, underscoring the higher quality of the MMC data — which, despite being sampled for Swahili and smaller in size for Burmese, still led to markedly better performance.

6 Conclusion

In this work, we compile translation data created by translators working for The Church of Jesus Christ of Latter-day Saints and present it in the Massively Multi-way-aligned Multilingual Corpus

(MMC). This corpus includes English-centric translation pairs into 98 languages, including both high and low resource languages from all over the world. When all of the translation data is aligned using the English source sentences as a pivot, the corpus includes 1.5 billion segment pairs in 9,702 language directions.

The quality of parallel translation segments in this corpus is showcased through three experiments. We demonstrate that finetuning NLLB on 80 languages already supported by the model using this corpus improves 57.7% of the 6320 possible language directions by 0.5 to 10.2 chrF++ points, even in an out-of-domain setting. We also showcase the quality of data for the 18 languages not supported by NLLB or FLORES by finetuning an NLLB model on just those languages, releasing the model weights for future research and use.

Additionally, we compare MMC single-language pair data to Paracrawl data and show that the quality of MMC yields improvements beyond using Paracrawl to finetune NLLB, especially for lower-resource languages.

This corpus can advance NMT and Large Language Model (LLM) research by providing low-resource language training data for improved MT models, including the first MT models trained on authentic, human-quality translation pairs for some languages. We anticipate widespread use of this data for accelerating global language technology development. The anticipated licensing of this data will make it available for training translation and language models in both research and commercial contexts.

Limitations

Due to the amount of data and potential flaws in the automatic cleaning of the corpus, there may still be errors in spelling and character encodings. We have

performed many manual spot-checks of the data, but incomplete sentences, misalignments, or other problems may still exist. Our experiments cover a limited set of multilingual finetuning scenarios and may not generalize to all model sizes or language combinations.

Ethics Statement

As with any religious texts, the corpus contains scriptures, doctrines, and teachings of the Church of Jesus Christ of Latter-day Saints, with which people of differing faiths and belief systems may disagree. The Church respects and supports the rights of others to believe as they will, as stated in its scriptural canon: "We claim the privilege of worshipping Almighty God according to the dictates of our own conscience, and allow all men the same privilege, let them worship how, where, or what they may."¹³

Some names of individuals and other limited information about them (but not what is normally considered personally identifiable information, or PII) are included in the corpus if the information is publicly available on the Church's website. However, efforts have been taken to remove such references, except in cases of widely known general and regional Church leadership, both past and present.

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¹³Articles of Faith 1:11

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A Cleaning Steps

We apply the following cleaning steps in our data cleaning pipeline:

1. Remove pairs containing empty source or target segments.
2. Remove pairs when the source segment exactly or nearly matches the target segment.
3. Remove duplicate source-target pairs.
4. Remove pairs with segments containing mostly non-alphabetic characters.
5. Remove pairs with segments containing abnormally long sequences of characters without spaces, including segments that are only URLs.
6. Remove pairs containing segments with unbalanced brackets.
7. Remove pairs containing fewer than 3 words in the English source segment.
8. Remove pairs with segments containing a higher number of characters than 5 standard deviations above the mean for that language (sentences that are too long).
9. Remove pairs in which the ratio of the lengths of the source and target segments exceeds a certain cutoff.
10. Normalize escaped Unicode characters.
11. Validate and normalize character encodings for each language.
12. Normalize whitespace
13. Shorten sequences of excessively repeated punctuation.
14. Normalize quotation marks.
15. Normalize HTML entities.
16. Remove all markup tags.

B Dataset Statistics

Table 7 shows the information about each of the target languages within the corpus, including the ISO 639-3 code, language name, language family according to Glottolog (<https://glottolog.org/>), language orthographic script, and the count of English $\rightarrow$ X translation segments in the corpus.

The heat map in Figure 3 shows the number of segments that can be pivoted through English to form parallel segments for other pairs of languages in the dataset.

C More Results

Table 8 showcases the average BLEU and chrF++ scores after finetuning NLLB on the 18 unsupported languages, aggregated by Source language and then by Target language.

Table 9 showcases all BLEU scores between each source and target language for the 18 languages unsupported by NLLB and FLORES in addition to English: Bislama, Chuukese, Efik, Fante, Kiribati, Hindi (Fiji), Hiligaynon, Hmong, Kekchi, Kosraean, Marshallese, Navajo, Palauan, Pohnpeian, Quichua (Ecuador), Tahitian, Tongan, and Yapese.

Figure 4 contains the differences in spBLEU comparing the baseline and finetuned NLLB model. We observe more outliers for spBLEU scores than for chrF++ scores, likely due to the pitfalls of BLEU for morphologically rich languages.

Finetuning NLLB on the 80 previously supported languages took approximately 4.5 hours using 8 A100 GPUs. Finetuning NLLB to support the 18 new languages took approximately 15 hours on 8 A100 GPUs. Inference for both models ranges from 5-13 sentences per second on a single A100 GPU.

ISO 639-3	Language	Family	Script	Count	FLORES Support
afr	Afrikaans	Indo-European	Latin	70940	Yes
amh	Amharic	Afro-Asiatic	Ge'ez	122892	Yes
ara	Arabic	Afro-Asiatic	Arabic	131000	Yes
bis	Bislama	Indo-European	Latin	273534	No
bul	Bulgarian	Indo-European	Cyrillic	662725	Yes
ceb	Cebuano	Austronesian	Latin	1125493	Yes
ces	Czech	Indo-European	Latin	724051	Yes
chk	Chuukese	Austronesian	Latin	80249	No
dan	Danish	Indo-European	Latin	983493	Yes
deu	German	Indo-European	Latin	1546154	Yes
efi	Efik	Atlantic-Congo	Latin	31376	No
ell	Greek	Indo-European	Greek	313665	Yes
est	Estonian	Uralic	Latin	608237	Yes
fat	Fante	Atlantic-Congo	Latin	78651	No
fij	Fijian	Austronesian	Latin	543683	Yes
fin	Finnish	Uralic	Latin	937551	Yes
fra	French	Indo-European	Latin	1931416	Yes
gil	Kiribati	Austronesian	Latin	339303	No
grn	Guarani	Tupian	Latin	78045	Yes
hat	Haitian	Indo-European	Latin	264359	Yes
hif	Hindi (Fiji)	Indo-European	Latin	63974	No
hil	Hiligaynon	Austronesian	Latin	111052	No
hin	Hindi	Indo-European	Devanagari	90444	Yes
hrv	Croatian	Indo-European	Latin	641221	Yes
hun	Hungarian	Uralic	Latin	168582	Yes
hww	Hmong Daw	Hmong-Mien	Latin	132579	No
hye	Armenian	Indo-European	Armenian	666189	Yes
ibo	Igbo	Atlantic-Congo	Latin	70730	Yes
ilo	Ilokano	Austronesian	Latin	193388	Yes
ind	Indonesian	Austronesian	Latin	856556	Yes
isl	Icelandic	Indo-European	Latin	205238	Yes
ita	Italian	Indo-European	Latin	1774804	Yes
jpn	Japanese	Japonic	Japanese	1503092	Yes
kam	Kikamba	Atlantic-Congo	Latin	4739	Yes
kan	Kannada	Dravidian	Kannada	5878	Yes
kat	Georgian	Kartvelian	Georgian	78264	Yes
kaz	Kazakh	Turkic	Cyrillic	4496	Yes
kek	Kekchi	Mayan	Latin	181710	No
khm	Khmer	Austroasiatic	Khmer	810075	Yes
kin	Kinyarwanda	Atlantic-Congo	Latin	26313	Yes
kor	Korean	Koreanic	Hangul	1410219	Yes
kos	Kosraean	Austronesian	Latin	84610	No
lao	Lao	Tai-Kadai	Lao	130692	Yes
lav	Latvian	Indo-European	Latin	668816	Yes
lin	Lingala	Atlantic-Congo	Latin	111316	Yes
lit	Lithuanian	Indo-European	Latin	642082	Yes
mah	Marshallese	Austronesian	Latin	341097	No
mkd	Macedonian	Indo-European	Cyrillic	39180	Yes
mlg	Malagasy	Austronesian	Latin	729155	Yes

mlt	Maltese	Afro-Asiatic	Latin	74979	Yes
mon	Mongolian	Mongolic-Khitani	Cyrillic	925796	Yes
msa	Malay	Austronesian	Latin	275124	Yes
mya	Burmese	Sino-Tibetan	Myanmar	19770	Yes
nav	Navajo	Athabaskan-Eyak-Tlingit	Latin	30889	No
nld	Dutch	Indo-European	Latin	1005049	Yes
nob	Norwegian	Indo-European	Latin	1005118	Yes
npi	Nepali	Indo-European	Devanagari	42672	Yes
pap	Papiamentu	Indo-European	Latin	64855	Yes
pau	Palauan	Austronesian	Latin	50994	No
pes	Persian	Indo-European	Arabic	88258	Yes
pol	Polish	Indo-European	Latin	745965	Yes
pon	Pohnpeian	Austronesian	Latin	65791	No
por	Portuguese	Indo-European	Latin	2150090	Yes
por	Portuguese (Euro)	Indo-European	Latin	71432	Yes
que	Quichua (Ecuador)	Quechuan	Latin	18051	No
ron	Romanian	Indo-European	Latin	797729	Yes
rus	Russian	Indo-European	Latin	1437848	Yes
sin	Sinhalese	Indo-European	Sinhala	64383	Yes
slk	Slovak	Indo-European	Latin	218399	Yes
slv	Slovenian	Indo-European	Latin	221788	Yes
smo	Samoan	Austronesian	Latin	1206051	Yes
sna	Shona	Atlantic-Congo	Latin	80868	Yes
sot	South Sotho	Atlantic-Congo	Latin	21256	Yes
spa	Spanish	Indo-European	Latin	2292344	Yes
sqi	Albanian	Indo-European	Latin	771789	Yes
srp	Serbian (Cyrillic)	Indo-European	Cyrillic	174296	Yes
swe	Swedish	Indo-European	Latin	959704	Yes
swh	Swahili	Atlantic-Congo	Latin	346327	Yes
tah	Tahitian	Austronesian	Latin	406538	No
tam	Tamil	Dravidian	Tamil	74068	Yes
tel	Telugu	Dravidian	Telugu	74886	Yes
tgl	Tagalog	Austronesian	Latin	1155293	Yes
tha	Thai	Tai-Kadai	Thai	906820	Yes
ton	Tongan	Austronesian	Latin	1191802	No
tpi	Tok Pisin	Indo-European	Latin	31191	Yes
tsn	Tswana	Atlantic-Congo	Latin	49989	Yes
tur	Turkish	Turkic	Latin	114528	Yes
twi	Twi	Atlantic-Congo	Latin	69340	Yes
ukr	Ukrainian	Indo-European	Cyrillic	885055	Yes
urd	Urdu	Indo-European	Arabic	98692	Yes
vie	Vietnamese	Austroasiatic	Latin	633954	Yes
xho	Xhosa	Atlantic-Congo	Latin	87180	Yes
yap	Yapese	Austronesian	Latin	62737	No
yor	Yoruba	Atlantic-Congo	Latin	75874	Yes
yue	Cantonese	Sino-Tibetan	Han (Trad.)	109790	Yes
zho	Chinese S	Sino-Tibetan	Han (Simp.)	706146	Yes
zho	Chinese T	Sino-Tibetan	Han (Trad.)	1162819	Yes
zul	Zulu	Atlantic-Congo	Latin	86410	Yes

Table 7: Information on each language included the corpus. We include ISO code, language, language family, orthographic script, and count of English-X parallel segments in the corpus.

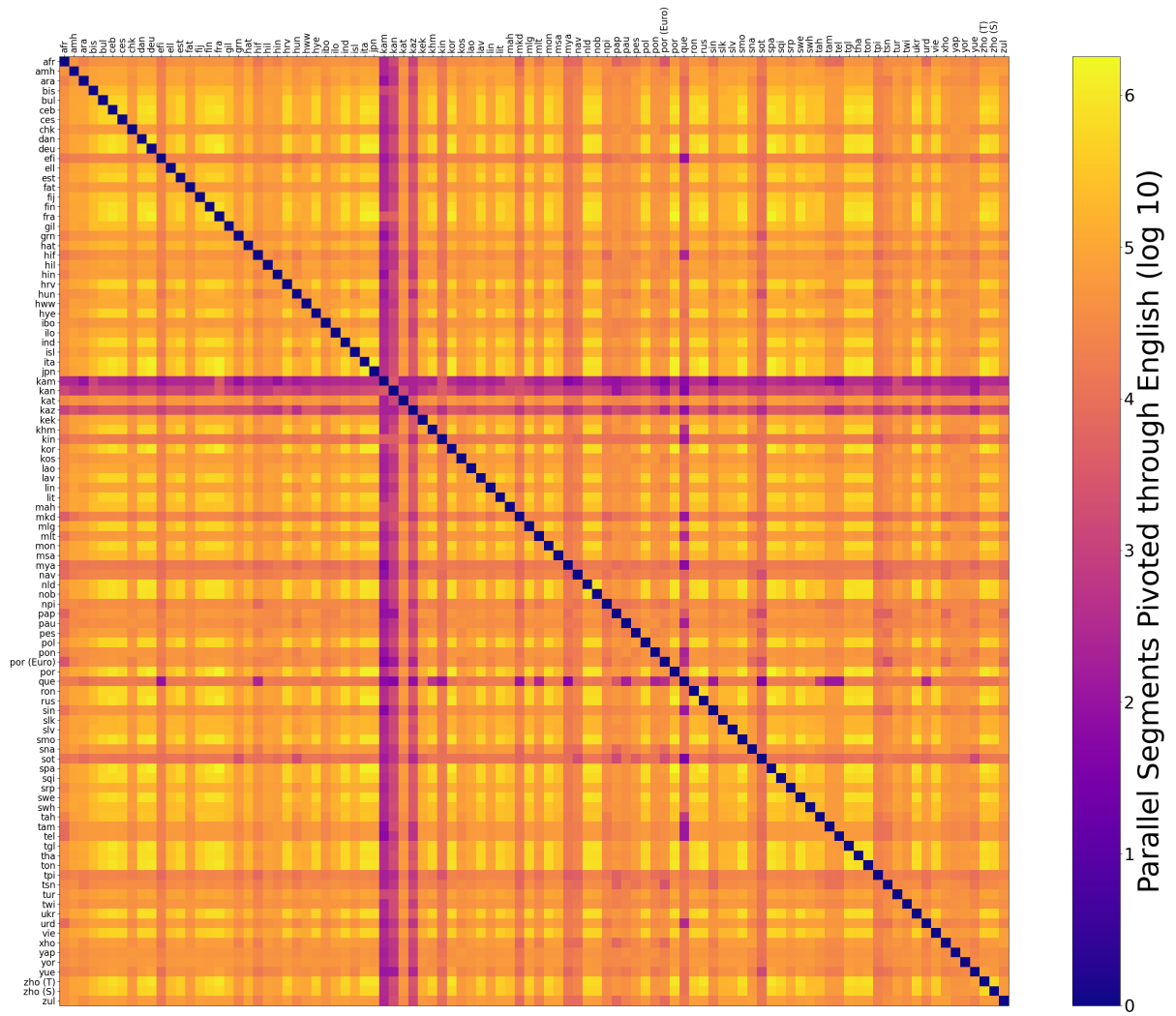


Figure 3: The number of parallel segments per language pair given (depicted by ISO codes) when pivoted through English. Note that most language pairs have between 10-100k parallel segments.

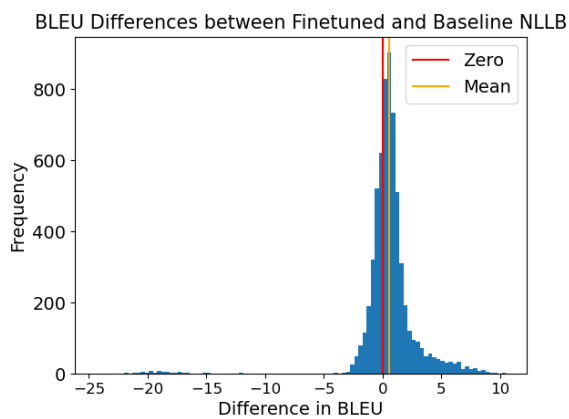


Figure 4: Histogram showing the differences between the spBLEU scores for the baseline and finetuned models. Positive values indicate a higher spBLEU score for the finetuned model. Zero difference and the mean difference are indicated by the red and orange lines respectively.

Direction	BLEU	chrF++	Direction	BLEU	chrF++
bis → X	26.88	50.32	X → bis	35.01	55.60
chk → X	26.31	48.61	X → chk	25.43	49.26
efi → X	25.39	47.78	X → efi	19.69	44.61
eng → X	31.65	55.28	X → eng	43.17	59.09
fat → X	27.06	50.05	X → fat	24.93	47.96
gil → X	25.54	48.02	X → gil	27.13	51.76
hif → X	27.62	50.50	X → hif	14.18	39.47
hil → X	29.13	51.68	X → hil	32.93	56.21
hww → X	23.54	46.16	X → hww	36.03	54.33
kek → X	23.78	46.37	X → kek	27.16	50.34
kos → X	25.71	48.38	X → kos	28.35	51.11
mah → X	27.32	50.08	X → mah	28.52	50.31
nav → X	13.31	34.12	X → nav	4.69	25.11
pau → X	23.62	46.13	X → pau	26.63	50.06
pon → X	24.70	46.98	X → pon	19.91	45.90
tah → X	26.55	49.60	X → tah	18.73	41.25
ton → X	27.08	50.14	X → ton	33.10	54.25
yap → X	22.91	44.98	X → yap	13.25	38.62

Table 8: Aggregated BLEU and chrF++ score results after finetuning NLLB on the 18 unsupported languages by source and target language. Quichua (que) is not included here due to only having English-centric test sets.

tgt src	bis	chk	efi	eng	fat	gil	hif	hil	hww	kek	kos	mah	nav	pau	pon	que	tah	ton	yap
bis	-	26.80	20.88	48.76	27.49	29.08	14.73	35.43	38.95	29.42	30.64	31.79	4.57	27.74	20.96	-	20.10	35.96	13.57
chk	35.43	-	20.61	44.29	24.75	28.38	14.93	33.17	36.75	28.17	29.99	30.49	4.91	27.26	21.28	-	19.21	34.05	13.67
efi	34.47	24.93	-	41.64	24.99	26.21	13.99	32.46	35.04	26.70	27.81	28.26	4.74	26.59	19.42	-	18.59	32.73	13.08
eng	46.79	32.84	25.69	-	34.14	33.54	19.93	46.45	43.46	32.50	35.93	38.69	5.08	30.83	24.94	37.30	22.91	43.14	15.62
fat	38.16	26.19	20.99	46.25	-	28.67	15.10	35.67	38.74	28.24	29.39	30.08	4.81	27.81	20.67	-	19.99	35.63	13.64
gil	34.81	25.94	19.96	42.21	25.13	-	14.21	33.22	36.15	27.18	28.96	29.09	4.73	27.40	20.45	-	18.63	33.01	13.14
hif	37.81	26.91	20.53	50.10	26.76	28.16	-	35.02	37.90	27.45	28.88	29.55	4.65	27.17	20.33	-	19.91	34.99	13.37
hil	41.07	29.07	22.95	55.41	30.14	31.64	16.56	-	40.61	30.63	32.09	33.50	4.99	29.25	22.56	-	21.50	38.47	14.70
hww	33.22	23.96	18.34	38.19	23.24	26.37	13.47	30.90	-	26.81	27.19	26.34	4.90	25.89	19.16	-	17.40	31.79	12.94
kek	32.78	23.83	17.74	37.35	22.87	25.35	13.04	31.43	35.70	-	26.52	26.46	4.37	25.61	18.47	-	18.24	31.48	12.96
kos	35.37	26.44	20.47	43.04	24.84	27.94	14.34	33.34	36.19	27.20	-	29.20	4.46	27.02	20.78	-	18.96	33.39	14.06
mah	38.38	27.95	21.55	48.05	27.40	29.03	15.29	35.82	37.76	28.42	31.02	-	4.90	27.89	21.67	-	19.98	35.65	13.76
nav	16.46	12.74	9.69	17.54	11.70	13.53	7.25	15.53	19.06	15.11	14.08	13.00	-	16.62	9.61	-	10.06	16.31	8.03
pau	32.14	24.53	18.36	36.80	22.84	25.02	13.06	30.34	33.81	26.05	26.84	26.17	4.63	-	19.13	-	17.80	30.95	13.06
pon	32.93	24.99	18.70	39.42	23.16	26.71	13.67	31.06	34.63	26.00	27.55	27.21	4.58	26.36	-	-	18.24	31.96	12.83
que	-	-	-	61.28	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
tah	36.76	25.46	19.87	44.61	25.96	27.22	14.16	35.16	37.07	28.12	29.00	29.55	4.38	26.49	19.86	-	-	34.86	12.85
ton	38.53	27.08	21.00	48.18	27.86	29.65	14.91	36.48	38.52	28.95	30.31	31.20	4.68	27.80	21.12	-	20.14	-	13.94
yap	30.05	22.67	17.39	33.92	20.57	24.68	12.37	28.28	32.25	24.84	25.71	24.32	4.35	24.96	18.03	-	16.71	28.41	-

Table 9: BLEU scores for the NLLB model finetuned on the 18 previously unsupported languages for every test set language direction.